

Permutations

What does it mean?

Permutations count the number of ways to **arrange** items where **order matters**.

Key Ideas:

- Order is important
- ABC is different from BAC
- Used for arrangements, rankings, positions
- Keywords: "arrange", "order", "first, second, third", "sequence"

Formula:

- $P(n,r) = n!/(n-r)!$
- n = total items
- r = items being arranged

Combinations

What does it mean?

Combinations count the number of ways to **select** items where **order doesn't matter**.

Key Ideas:

- Order is not important
- ABC is the same as BAC
- Used for selections, teams, groups
- Keywords: "choose", "select", "pick", "committee"

Formula:

- $C(n,r) = n!/(r!(n-r)!)$
- n = total items
- r = items being selected

Problem 1:

How many ways can you arrange the letters A, B, C?

Solve:

Total letters = 3

Arrange all letters = $P(3,3) = 3! = 6$

Detailed Explanation:

- We want to arrange ALL 3 letters
- Order matters ($ABC \neq BAC$)
- Use permutation formula: $P(3,3) = 3!$
- Answer: 6 ways
- All arrangements: ABC, ACB, BAC, BCA, CAB, CBA

Problem 2:

From 5 students, how many ways can you choose 3 for a team?

Solve:

Total students = 5

Choose 3 students = $C(5,3)$

$C(5,3) = 10$

Detailed Explanation:

- We want to SELECT 3 students from 5
- Order doesn't matter (team of Alice, Bob, Carol = team of Carol, Bob, Alice)
- Use combination formula: $C(5,3)$
- Answer: 10 ways

Problem 3:

How many ways can 4 people sit in a row?

Solve:

Total people = 4

Arrange all people = $P(4,4) = 24$

Detailed Explanation:

- We want to arrange ALL 4 people in seats
- Order matters (who sits in which position)
- First seat: 4 choices
- Second seat: 3 choices
- Third seat: 2 choices
- Fourth seat: 1 choice
- Total: $P(4,4) = 24$ ways

Problem 4:

From the word "MATH", how many 2-letter arrangements can you make?

Solve:

Total letters = 4 (M, A, T, H)

Arrange 2 letters = $P(4,2) = 12$

Detailed Explanation:

- We want to arrange 2 letters from 4 available
- Order matters ($MA \neq AM$)
- Use permutation formula: $P(4,2)$
- First position: 4 choices
- Second position: 3 choices (one letter already used)
- Total: $4 \times 3 = 12$ arrangements

Problem 5:

How many ways can you choose 2 colors from red, blue, green, yellow?

Solve:

Total colors = 4

Choose 2 colors = $C(4,2) = 6$

Detailed Explanation:

- We want to SELECT 2 colors from 4 available
- Order doesn't matter (red+blue = blue+red)
- Use combination formula: $C(4,2)$
- Answer: 6 ways
- All combinations: (red,blue), (red,green), (red,yellow), (blue,green), (blue,yellow), (green,yellow)

Problem 6:

A class has 10 students. How many ways can you choose a president, vice president, and secretary?

Solve:

Total students = 10

Choose and arrange 3 positions = $P(10,3) = 720$

Detailed Explanation:

- We need 3 different positions: president, vice president, secretary
- Order matters (different positions = different arrangements)
- Use permutation: $P(10,3)$
- President: 10 choices
- Vice President: 9 choices (can't be president)
- Secretary: 8 choices (can't be president or VP)
- Total: $10 \times 9 \times 8 = 720$ ways

Problem 7:

From 8 books, how many ways can you select 5 books for your shelf?

Solve:

Total books = 8

Select 5 books = $C(8,5) = 56$

Detailed Explanation:

- We want to SELECT 5 books from 8 available
- Order doesn't matter (just selecting books, not arranging them)
- Use combination formula: $C(8,5)$
- Answer: 56 ways

Problem 8:

How many 4-digit numbers can be formed using digits 1,2,3,4,5,6 without repetition?

Solve:

Total digits = 6 (1,2,3,4,5,6)

Arrange 4 digits = $P(6,4) = 360$

Detailed Explanation:

- We want to form 4-digit numbers using 4 different digits from 6 available
- Order matters ($1234 \neq 4321$, different numbers)
- No repetition means each digit used only once
- Use permutation: $P(6,4)$
- First digit: 6 choices
- Second digit: 5 choices
- Third digit: 4 choices
- Fourth digit: 3 choices
- Total: $6 \times 5 \times 4 \times 3 = 360$ numbers

Problem 9:

A pizza shop offers 10 toppings. How many ways can you choose exactly 3 toppings?

Solve:

Total toppings = 10

Choose 3 toppings = $C(10,3) = 120$

Detailed Explanation:

- We want to SELECT exactly 3 toppings from 10 available
- Order doesn't matter (pepperoni+mushroom+cheese = cheese+pepperoni+mushroom)
- Use combination formula: $C(10,3)$
- Answer: 120 ways

Problem 10:

How many ways can 6 friends sit around a circular table?

Solve:

For circular arrangements = $(n-1)!$

6 friends around circular table = $(6-1)! = 5! = 120$

Detailed Explanation:

- Circular arrangements are different from linear arrangements
- In a circle, rotations are considered the same
- Fix one person's position, then arrange the remaining $(n-1)$ people
- Formula for circular permutations: $(n-1)!$
- Fix one friend in a seat, arrange remaining 5 friends = $5! = 120$ ways

Problem 11:

How many ways can the letters of the word "BOOK" be arranged?

Solve:

Total letters = 4

Repeated letter O appears 2 times

Arrangements = $4!/2! = 24/2 = 12$

Detailed Explanation:

- BOOK has 4 letters: B, O, O, K
- Letter O is repeated 2 times
- For repeated items, use formula: $n!/r_1! \times r_2! \times \dots$
- Where r_1, r_2 are repetition counts
- Arrangements = $4!/2! = 12$ ways
- Examples: BOOK, BOKO, BKO O, OBKO, etc.

Problem 12:

A committee of 5 people is to be formed from 7 men and 4 women. How many ways can this be done if the committee must have exactly 3 men and 2 women?

Solve:

Choose 3 men from 7 men = $C(7,3) = 35$

Choose 2 women from 4 women = $C(4,2) = 6$

Total ways = $C(7,3) \times C(4,2) = 35 \times 6 = 210$

Detailed Explanation:

- We need exactly 3 men AND exactly 2 women
- This is a two-step selection process
- Step 1: Choose 3 men from 7 available men = $C(7,3) = 35$ ways, order doesn't matter
- Step 2: Choose 2 women from 4 available women = $C(4,2) = 6$ ways, order doesn't matter
- Since we need both steps, multiply the results = $35 \times 6 = 210$ ways (Product Rule)

Problem 13:

How many ways can 8 people be divided into 2 groups of 4 each?

Solve:

Total ways to choose 4 people from 8 = $C(8,4) = 70$

Since groups are indistinguishable, divide by $2!$

Final answer = $C(8,4)/2! = 70/2 = 35$

Detailed Explanation:

- First, choose 4 people from 8 for the first group = $C(8,4) = 70$
- The remaining 4 automatically form the second group
- But the two groups are identical (Group A and Group B are the same as Group B and Group A)
- So we divide by $2!$ to avoid double counting
- Answer: $70/2 = 35$ ways

Problem 14:

How many 5-letter words can be formed from the letters of "APPLE" where all letters are used?

Solve:

Total letters = 5 (A, P, P, L, E)

Letter P appears 2 times

Arrangements = $5!/2! = 60$

Detailed Explanation:

- APPLE has 5 letters: A, P, P, L, E
- Letter P is repeated 2 times
- Use formula for repeated items: $n!/r_1! \times r_2! \times \dots$
- Where r_1, r_2 are repetition counts
- Arrangements = $5!/2! = 60$ words
- Examples: APPLE, APEPL, ALPEP, PALPE, etc.

Problem 15:

A student must answer exactly 8 questions out of 12 questions on an exam. The first 5 questions are compulsory, and the student can choose any 3 from the remaining 7 questions. How many ways can the student select the questions?

Solve:

Compulsory questions = 5

Remaining questions = $12 - 5 = 7$

Choose 3 from remaining 7 = $C(7,3) = 35$

Detailed Explanation:

- The first 5 questions are compulsory (no choice here)
- From the remaining 7 questions, choose exactly 3, order doesn't matter
- This is a simple combination problem: $C(7,3)$
- Answer: 35 ways to select the optional questions

Problem 16:

How many ways can you arrange 10 people in a row if 3 specific people must sit together?

Solve:

Treat the 3 specific people as ONE unit

Total units to arrange = $10 - 3 + 1 = 8$ units

Arrange 8 units = $8!$ ways

Arrange the 3 people within their unit = $3!$ ways

Total arrangements = $8! \times 3! = 241,920$

Detailed Explanation:

- Group the 3 specific people as one group
- Now we have 8 units to arrange (7 individual people + 1 group of 3)
- Arrange these 8 units = $8! = 40,320$ ways
- Within the group, the 3 people can be arranged = $3! = 6$ ways
- Total arrangements = $8! \times 3! = 241,920$ ways

Problem 17:

In how many ways can 6 boys and 4 girls be seated in a row so that no two girls sit together?

Solve:

Step 1:

Arrange 6 boys = $6! = 720$ ways

Step 2:

This creates 7 spaces for girls (before first boy, between boys, after last boy) _ B _ B _ B _ B _ B _

Step 3:

Choose 4 spaces from 7 for girls = $C(7,4) = 35$ ways

Step 4:

Arrange 4 girls in chosen spaces = $4! = 24$ ways

Total = $6! \times C(7,4) \times 4! = 720 \times 35 \times 24 = 604,800$

Detailed Explanation:

- First arrange all 6 boys in a row = $6!$ ways
- This creates 7 possible positions for girls (marked with _)
- Choose 4 positions from these 7 for the 4 girls = $C(7,4)$ ways
- Arrange the 4 girls in the chosen positions = $4!$ ways
- Total = $720 \times 35 \times 24 = 604,800$ ways

Problem 18:

A bag contains 5 red balls, 3 blue balls, and 2 green balls. How many ways can you select 6 balls such that you get at least 2 balls of each color?

Solve:

Minimum requirement: 2 red, 2 blue, 2 green = 6 balls total

This exactly matches our selection of 6 balls

So we need EXACTLY 2 of each color.

Ways to choose 2 red from 5 = $C(5,2) = 10$

Ways to choose 2 blue from 3 = $C(3,2) = 3$

Ways to choose 2 green from 2 = $C(2,2) = 1$

Total ways = $C(5,2) \times C(3,2) \times C(2,2) = 10 \times 3 \times 1 = 30$

Detailed Explanation:

- We need to select exactly 6 balls with at least 2 of each color
- Since we need at least 2 red, 2 blue, 2 green, and that totals 6 balls
- We need EXACTLY 2 of each color (no more, no less)
- Choose 2 red from 5 red balls = $C(5,2) = 10$ ways
- Choose 2 blue from 3 blue balls = $C(3,2) = 3$ ways
- Choose 2 green from 2 green balls = $C(2,2) = 1$ way
- Total: $10 \times 3 \times 1 = 30$ ways